In a Different Voice

Gender Composition and Scientific Writing in Economics

Alessandra Cassar   Austin Coffelt   Sopiko Datuashvili   Robizon Khubulashvili

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Scientific writing is not a neutral container for ideas

How research is framed shapes evaluation — dominant standards may systematically overlook a "different voice" (Gilligan 1993)

- Does gender composition predict systematic differences in how economists write?
- Builds on Hengel (2022) *Publishing While Female*: female-authored papers more readable; writing gap widens during peer review
- Extends to **15 additional** tonal dimensions — assertiveness, hedging, directness, novelty claims, and more

Preview of findings

- Clarity gap: female-majority teams score higher on readability, directness, and evidentiary support; lower on jargon
- Tonal dimensions — assertiveness, hedging, emotional valence — no systematic difference

This paper contributes to three literatures

- **Gender, voice, & communication** — Gilligan (1993): dominant evaluative standards may systematically overlook a "different voice"
- **Gender and economics outcomes** — Card et al. (2020): referees treat submissions differently by gender; Sarsons (2017): collaborative work differentially credited
- **Readability and peer review** — Hengel (2022) *Publishing While Female*: female-authored papers more readable; writing gap widens during review
- **Our contribution** — extend Hengel with 15 additional tonal dimensions; sharpen the "different voice" debate by identifying *which* dimensions move with team composition

9,117 abstracts from four top economics journals, 1950–2015

- Source: Hengel (2022) replication dataset — 4 journals, 1950–2015; analysis on abstracts only
- **Article-level controls**: JEL codes, submission & acceptance dates, citation count, pre-computed readability scores, blind-review indicator
- **Author-level controls**: gender, native language, institution, seniority, prominence (most-published co-author)
- Sample: 7,939 all-male articles, 319 all-female; $<10\%$ female authors overall
- Six authorship measures: female ratio, majority female, any female, 100% female, solo female, senior female

16-item LLM rubric: surface-level linguistic features only

Each criterion scored 1–10. No author intent or personality inferred.

G1 Creativity & Hedging

Modal Verb, Hedging, Qualifier Density, Ack. of Limitations, Caution

G2 Assertiveness & Voice

Assertiveness, Active/Passive Voice

G3 Structural Directness

Sentence Directness, Imperative-Form Occurrence

G4 Authorial Stance & Novelty

Pronoun Commitment, Novelty Claims, Jargon Density, Emotional Valence

G5 Support & Impact

Evidence & Citation Usage, Practical Orientation

Standalone Readability

† Jargon/Technicality Density is scale-flipped ($11-\text{raw}$) so that high score $=$ less jargon.

GPT-5 with chain-of-thought reasoning scored all 9,117 abstracts

- **Model:** GPT-5 — extended reasoning before answering improves accuracy on complex tasks (Shojaee et al. 2025)
- **Pipeline:** abstract $\rightarrow$ rubric $\rightarrow$ model $\rightarrow$ structured 16-score output
- **Prompt engineering:** step-by-step reasoning induced per criterion; shorter inputs increase accuracy (Modarressi et al. 2025)
- **Validation:** human-in-the-loop on small subset — audited outputs and iteratively improved prompt; then benchmarked against Hengel (2022) readability scores
- *Caveat:* some randomness in LLM outputs; reported SEs may understate true uncertainty

Three-step analysis: validate → mean differences → controlled regressions

(1) Benchmark LLM readability against Hengel (2022) measures **(2)** Unconditional mean differences across all 16 criteria **(3)** Controlled regressions

$$R_j = \beta_0 + \beta_1 \cdot \textit{FemaleRatio}_j + \gamma X_j + \delta Y_j + \varepsilon_j$$

X_j: journal×year FE, editor controls, blind-review indicator, JEL codes, theory/empirical flag

Y_j: institution FE, author prominence (max T FE), author seniority (max t FE), native-speaker indicator

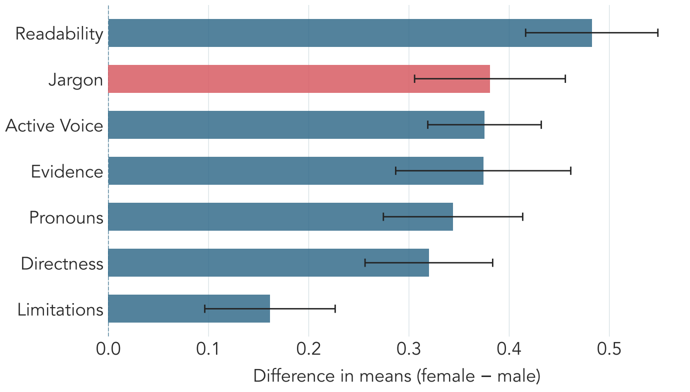
Nine specifications. SEs clustered on editor. Alternative authorship measures: majority-female binary, any-female binary, 100% female, solo, senior female.

LLM readability replicates Hengel's established pattern across all nine specifications

Readability measure	(4)	(5)
Flesch Reading Ease	1.26^{**}	1.49^{***}
Flesch-Kincaid Grade Level	0.24^{*}	0.26^{*}
Gunning Fog Index	0.40^{**}	0.42^{***}
SMOG Index	0.27^{**}	0.29^{**}
Dale-Chall Score	0.08	0.09
LLM Readability	$\mathbf{0.40}^{***}$	$\mathbf{0.32}^{***}$
N	9,117	9,117

OLS coefficients on female ratio. SEs clustered on editor. Col. (4): journal×year, N_j, institution, editor, blind-review. Col. (5) adds quality controls, native speaker, JEL, theory/empirical.

Female-majority teams score higher on clarity; tonal differences are negligible



Null: assertiveness, hedging, emotional valence — no difference between male- and female-majority teams.

Controlled regressions confirm the clarity differential; tonal dimensions remain null

Criterion	(4)	(5)
Jargon/Technicality Density†	0.55^{***}	0.48^{***}
Sentence Length & Directness	0.31^{***}	0.25^{**}
Evidence & Citation Usage	0.46^{***}	0.36^{***}
Pronoun Commitment	0.39^{**}	0.32
Acknowledgement of Limitations	0.24^{**}	0.15
Active/Passive Voice	0.16	0.15
N	9,117	9,117

OLS coefficients on female ratio. SEs clustered on editor. Col. (4): journal×year, N_j, institution, editor, blind-review. Col. (5) adds quality controls, native speaker. †Scale-flipped (11−raw); positive = less jargon.

Clarity differential is robust across six authorship measures

	Authorship measure					
Criterion	Female ratio	$\geq$50% female	$\geq$1 female	100% female	Solo female	Senior female
LLM Readability	0.32^{***}	0.22^{***}	0.18^{***}	0.31^{***}	0.26^{**}	0.33^{***}
Jargon/Technicality†	0.48^{***}	0.31^{***}	0.28^{***}	0.51^{***}	0.45^{***}	0.57^{***}
Sentence Directness	0.25^{**}	0.20^{**}	0.17^{***}	0.21^{*}	0.16	0.21^{**}
Assertiveness	-0.05	-0.00	-0.02	-0.13	-0.16	-0.14
Emotional Valence	-0.03	-0.02	-0.01	-0.04^{*}	-0.04^{**}	-0.03^{*}
N	9,117					

OLS col (5) coefficients on authorship measure. SEs clustered on editor. †Scale-flipped (11−raw); positive = less jargon.

Female-majority teams in economics write more clearly.

LLM readability validates against Hengel (2022). In mean differences and controlled regressions, female-majority teams score higher on readability, directness, active voice, and evidentiary support, and lower on jargon — not more hesitant or emotional. Robust across control levels and authorship specifications; LLM measurement error not yet fully quantified.

Appendix: Summary statistics by gender composition

	All-Male ($N = 7{,}951$)	Mixed ($N = 853$)	All-Female ($N = 313$)	Full ($N = 9{,}117$)
Readability	6.70 (1.38)	7.42 (1.25)	7.37 (1.30)	6.79 (1.38)
Directness	5.94 (2.08)	6.71 (1.81)	6.53 (1.86)	6.03 (2.06)
Jargon (raw)	7.00 (1.73)	6.48 (1.86)	6.07 (1.92)	6.92 (1.76)
Evidence	1.68 (1.34)	2.31 (1.90)	2.19 (1.73)	1.75 (1.43)
Novelty	2.27 (2.12)	2.26 (2.20)	1.90 (1.85)	2.25 (2.12)
Assertiveness	8.18 (1.55)	8.43 (1.28)	8.04 (1.55)	8.20 (1.53)
Hedging	8.35 (1.98)	8.49 (1.73)	8.07 (2.11)	8.35 (1.96)
Emotional val.	1.12 (0.37)	1.11 (0.35)	1.10 (0.31)	1.12 (0.37)

Means and SDs. Scored 1–10. Descriptive only; no significance stars.

Appendix: Tonal criteria — controlled regressions (G1–G2)

Criterion	(4)	(5)
Modal Verb Strength	-0.32^{*}	-0.34^{*}
Hedging Frequency & Type	-0.20^{*}	-0.17
Qualifier Density	0.00	0.01
Caution-Signaling Connectors	0.08	0.08
Assertiveness & Voice	-0.06	-0.05
N	9,117	

OLS coefficients on female ratio. SEs clustered on editor. Col. (4): journal$\times$year, N_j, institution, editor, blind-review. Col. (5) adds quality controls, native speaker, JEL, theory/empirical.

Appendix: Remaining criteria — controlled regressions (G3–G5)

Criterion	(4)	(5)
Imperative-Form Occurrence	-0.02^{***}	-0.02^{***}
Novelty-Claim Strength	-0.31^{***}	-0.30^{***}
Emotional Valence	-0.03	-0.03
Practical/Impact Orientation	0.09	0.05
N	9,117	

OLS coefficients on female ratio. SEs clustered on editor. Col. (4): journal×year, N_j, institution, editor, blind-review. Col. (5) adds quality controls, native speaker, JEL, theory/empirical.

Appendix: Tonal criteria — robustness across authorship specifications (G1–G2)

Criterion	Authorship measure: Female ratio	≥50% female	≥1 female	100% female	Solo female	Senior female
Modal Verb Strength	−0.34*	−0.14	−0.10	−0.55**	−0.59**	−0.59**
Hedging Frequency & Type	−0.17	−0.09	−0.09	−0.26*	−0.32**	−0.30**
Qualifier Density	0.01	0.02	−0.04	−0.03	−0.05	−0.05
Ack. of Limitations	0.15	0.10	0.12	0.20	0.10	0.14
Caution-Signaling	0.08	0.04	−0.02	0.03	0.10	0.01
Active/Passive Voice	0.15	0.12*	0.10*	0.13	0.25	0.12
N	9,117					

OLS col (5) coefficients on authorship measure. SEs clustered on editor.

Appendix: Remaining criteria — robustness across authorship specifications (G3–G5)

Criterion	Authorship measure					
	Female ratio	$\geq$50% female	$\geq$1 female	100% female	Solo female	Senior female
Imperative-Form Occurrence	-0.02^{***}	-0.01^{**}	-0.00	-0.02^{***}	-0.02^{**}	-0.02^{***}
Novelty-Claim Strength	-0.30^{***}	-0.21^{**}	-0.24^{**}	-0.28^{**}	-0.27^{*}	-0.33^{***}
Pronoun Commitment	0.32	0.39^{**}	0.15	-0.00	0.36^{*}	-0.05
Evidence & Citation Usage	0.36^{***}	0.30^{***}	0.25^{***}	0.26^{**}	0.21^{*}	0.29^{***}
Practical/Impact Orientation	0.05	0.04	0.02	0.01	0.02	0.07
N	9,117					

OLS col (5) coefficients on authorship measure. SEs clustered on editor.